

Learning Objectives

- Explain the client and server model and the difference between HTTP and HTTPS.
- Describe how HTTP requests and responses work, and explain the role of an API in exchanging data.

When you click a link or press “search,” your device does not hold the whole website — it asks a distant computer for it. Your browser is the client that makes the request; a server somewhere answers with a response. This exchange, governed by shared rules called protocols, is how every web application communicates. Knowing how requests, responses, and APIs work explains what happens between your click and the page you see.

? Guiding question: How do the client and server communicate to make a web application work, and what rules and messages do they use?

Key Fact:
Web applications work by a client sending requests and a server returning responses, using protocols (HTTP/HTTPS) and often APIs that exchange data in formats like JSON.

Key Concepts:

- client / server
- HTTP / HTTPS
- HTTP method
- status code
- API / JSON

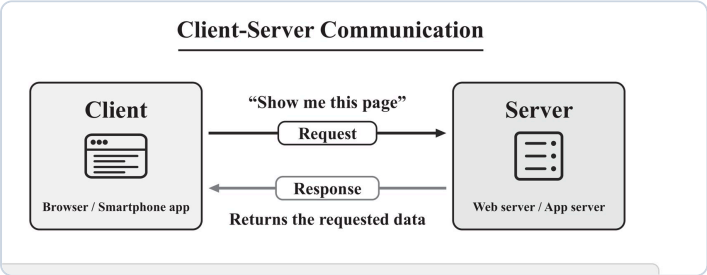
Point!

1. Client and Server

(1) **(Client)**…the side that requests data or processing from the server. Web browsers and smartphone apps fall into this category.

(2) **(Server)**…the side that receives requests from clients, processes them, and returns responses.

(3) A web application operates by the client sending a request and the server returning a response.



Client-Server Communication

KEY TERMS
client — the side that requests data or processing.
server — the side that processes requests and returns responses.



Many clients request; one server responds.

2. HTTP and HTTPS

(1) **(HTTP (HyperText Transfer Protocol))**…a communication protocol for exchanging data between clients (browsers and apps) and servers.

(2) **(HTTPS (HTTP Secure))**…a communication protocol that adds TLS encryption to HTTP. Because communication content is encrypted, the risk of its being read by third parties is reduced.

(3) Websites whose URLs begin with “https://” use HTTPS communication.

3. HTTP Requests and Responses

(1) Requests sent from the client to the server include information called an HTTP method that indicates “what kind of operation is desired.”

PAUSE & THINK

Why should a login page use HTTPS rather than plain HTTP?

Method	Role	Examples
GET	Retrieve data from the server	Displaying a web page, retrieving search results
POST	Send data to the server	Sending login information, submitting form contents

(2) Responses returned from the server to the client include a number indicating the result of processing, called a status code. Some typical examples are as follows.

Status code	Meaning
200	Success (the request was processed successfully)
404	Page not found (the specified URL does not exist)
500	Server error (a problem occurred on the server side)

4. API (Application Programming Interface)

(1) **(API)**…a system for exchanging data between different pieces of software.

(2) In web applications, the frontend often sends requests to the backend through an API and receives data.

(3) Data exchanged through an API is often described in a format called **(JSON (JavaScript Object Notation))**. JSON is a data format that is both easy for humans to read and for computers to process.

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A The client is the side that requests data or processing from the server, and the server is the side that returns the response.

B HTTPS is a communication protocol that adds encryption to HTTP.

C The GET method is for sending data to the server.

D Status code 404 indicates that the page is not found.

(2) Match each description (a–c) with the most appropriate term from the choices below (A–C).

a A communication protocol for exchanging data between web browsers and servers

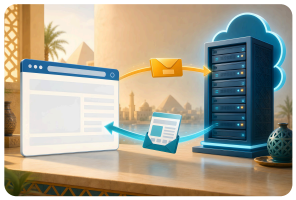
b A system for exchanging data between different pieces of software

c A communication protocol that adds encryption to HTTP

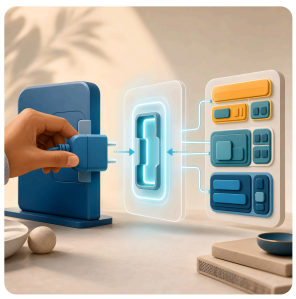
[Choices] **A** HTTP **B** API **C** HTTPS

KEY TERMS

HTTP method — says what is wanted (GET, POST).
status code — reports the result (200, 404, 500).



One HTTP request, one matching response.



An API is a fixed contract for exchanging data.

✓ Solution

(1)

A Correct description of the roles of client and server. Therefore, ○.

B HTTPS adds SSL/TLS encryption to HTTP. Therefore, ○.

C The GET method is for retrieving data from the server and the POST method is for sending data. Therefore, ×.

D 404 is a status code indicating that the page is not found. Therefore, ○.

(2)

a: The web communication protocol is HTTP. **A**

b: Data exchange between pieces of software is API. **B**

c: HTTP with added encryption is HTTPS. **C**

Try

1 Answer the following questions.

1. What is the term for the side that requests data or processing from the server (e.g., a web browser)?
2. What is the term for the side that receives requests from clients, processes them, and returns responses?
3. What is the term for the communication protocol that adds encryption to HTTP?
4. What is the term for the HTTP method for retrieving data from the server?
5. What is the term for the system for exchanging data between different pieces of software?

2 Answer the following questions.

(1) From the following options (A–D), choose the combination of status code and meaning that is NOT appropriate.

A 200 — The request was processed successfully

B 404 — Page not found

C 500 — The request was processed successfully

D 500 — A problem occurred on the server side

EXAM-STYLE QUESTION

Explain what happens, step by step, when a browser requests a web page and the server responds. **[6] Refer to:** client, request, HTTP method, response, status code.

PAUSE & THINK

If a browser receives status code 404, what should it tell the user — and whose side had the problem?

(2) From the following options (A–D), choose the one that most appropriately describes an API.

A A technology for adjusting the appearance of web pages

B A system for exchanging data between different pieces of software

C A technology for encrypting data and sending it safely

D An HTTP method for retrieving data from the server

⚙ Think as an Engineer — Investigate & Decide

A school in your city wants a web app where students log in, view their timetable, and download a PDF. You plan how the client and server communicate.

☐ **Send everything over plain HTTP**

☐ **Use HTTPS and the right method for each action**

1 · Choose the method. For three actions — loading a student's grade history, submitting the login form, and requesting today's schedule from another school system — say whether GET or POST fits, and for the last say whether an API is involved.

2 · Choose the protocol. Decide whether the login should use HTTP or HTTPS, and give the reason.

3 · Decide. Recommend the communication design and give two reasons. **In pairs,** compare your method choices and agree together on which action must use POST.

Give evidence for your answer.

Stuck? GET retrieves and shows; POST sends data such as a login; HTTPS encrypts so passwords are not intercepted.

🌍 In a New Context

A weather service offers an API that returns today's forecast when a client sends a request. Using what you learned, name which HTTP method the client would use to retrieve the forecast, why the service should use HTTPS, and one advantage of receiving the data as JSON.

KEY TERMS

API — lets different software exchange data.

JSON — a readable data format.

❓ Lesson Question — Answered

The lesson asked how the client and server communicate and what rules and messages they use. A web application works when a client — a browser or app — sends a request and a server processes it and returns a response. They communicate through a protocol: HTTP carries the messages, and HTTPS adds SSL/TLS encryption so the content is not easily intercepted. Each request carries an HTTP method that says what is wanted — GET to retrieve, POST to send — and each response carries a status code that reports the result, such as 200 for success or 404 for not found. Often the frontend and backend exchange data through an API, frequently in the JSON format, which is easy for both people and computers to read. Overall, communication in a web application is an organized exchange of requests and responses, shaped by protocols, methods, status codes, and APIs.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

A web application operates by communication between a (a), which requests data or processing from the server, and a (b), which receives the request, processes it, and returns a response. This communication uses a protocol called (c). The protocol that adds SSL/TLS encryption to (c) is called (d).

1. Fill in the blanks (a)–(d).

3 Answer the following questions.

(1) For each of operations (a–d), indicate which HTTP method is used: **A** GET, **B** POST.

a Displaying a web page

b Entering an ID and password into a login form and submitting them

c Displaying search results

d Submitting the contents of a contact form

(2) From the following options (A–D), choose the one that most appropriately describes HTTPS.

A A method for retrieving data from the server

B A system for exchanging data between different pieces of software

C A communication protocol that adds SSL/TLS encryption to HTTP

D A status code indicating that a problem has occurred on the server side

⭐ REFLECT & CHALLENGE

Reflect: When a page fails to load, which status code would help you most to understand why, and what would it tell you?
Challenge: Name one app you use and describe one request it sends and the response it expects.

⭐ Key takeaway

Remember:
Client requests, server responds — over HTTP/HTTPS, with a method (GET/POST), a status code, and often an API returning JSON.